

McPhase Explorer

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1 Introduction

McPhase Explorer is an IDE - like program based on the NetBeans platform (<http://www.platform.netbeans.org/>) designed to help users create, edit and view files associated with the *McPhase* program package.

Since the program is based on a well established platform, documentation to add and/or change the program's behaviour is readily available at <http://www.platform.netbeans.org/kb/>.

2 Architecture

The NetBeans platform uses a modular architecture. E.g. each file type support is capsuled in its own independent module which can be loaded in any NetBeans platform application. To get a quick overview of the architecture please visit <http://platform.netbeans.org/tutorials/nbm-quick-start.html>.

2.1 McPhaseBase

McPhaseBase is a module containing basic implementations of commonly used classes to streamline the development process of file type support modules. It provides

- support for multiple editors being associated with one file such as a plain text representation and a UI based representation,
- a basic implementation of a plain text editor (*SourceEditorBase*),
- a base class for visual editors (*VisualEditorBase*) including support for getting and setting key-value type variables.

2.2 Adding file type support

Adding editor support for a custom file type can be achieved easily by

1. adding a new module to the *McPhaseExplorer* application suite,
2. choosing *Module Development* → *File Type* in the new file wizard,
3. letting the newly created *DataObject* inherit from *org.mcphase.DataObjectBase* and implementing its abstract methods.

The only abstract method of *org.mcphase.base.DataObjectBase* is *DataEditorSupport createEditorSupport()* and returns an editor support for the file type. Custom editor supports can be provided (<http://bits.netbeans.org/dev/javadoc/org-openide-loaders/org-openide/text/DataEditorSupport.html>) or the helper class *org.mcphase.base.EditorSupportBase* can be used as shown in the following sample implementation

```
[...]
public DataEditorSupport createEditorSupport() {
    //Create a new editor support associated with this data object
    EditorSupportBase support = new EditorSupportBase(this);
    //Create a view description
    ViewBase visual = new ViewBase();
    //Set its display name
    visual.setDisplayName("some name");
    //Set its icon (extending Image)
    visual.setIcon(someIcon);
    //Set its editor (implementing MultiViewElement
    //often extending VisualEditorBase)
    visual.setViewElement(someEditor);
    //Create the view elements
    MultiViewDescription[] desc = {new SourceView(support), visual};
    //Add the view elements to the editor support
    support.setViewDescriptions(desc);
    //Set the default element
    support.setDefaultViewDescription(desc[0]);
    //return the editor support
    return support;
}
[...]
```

For a full implementation see the sample implementation of `mcphas.ini`. All classes used above are documented in either of the following javadoc documentations

NetBeans API List (<http://bits.netbeans.org/dev/javadoc/index.html>)

`DataEditorSupport`, `MultiViewDescription`

McPhaseBase Documentation (provided with this document) `EditorSupportBase`, `SourceView`, `ViewBase`, `VisualEditorBase`

3 Current development status

Internal support for the following file types is provided

sipf Single ion parameter file

ini McPhase algorithm control file

jpg, jpeg, png, gif, bmp Image files

ant Ant build scripts

All other file types are opened with the native OS application.

4 McPhaseBase Documentation

4.1 `org.mcphase.base.DataObjectBase`

The following is a brief summary of the most commonly used classes and their interfaces. A full documentation can be found in the javadoc provided with this document. *`org.mcphase.base.DataObjectBase`* represents a file object and provides support for multiple editors being associated with it.

It has one abstract method, *`DataEditorSupport createEditorSupport()`*, which should be overridden in derived classes to provide an editor support. The helper class *`org.mcphase.base.EditorSupportBase`* may be helpful in doing so.

4.2 `org.mcphase.base.EditorSupportBase`

`org.mcphase.base.EditorSupportBase` implements a basic editor support to be used with multiple editors.

It requires a `DataObject` to be passed to its constructor and provides the following functionality

MultiViewDescription **getViewDescriptions()** Gets the view descriptions representing the various editor associated with the editor support.

void setViewDescriptions(MultiViewDescription **viewDescriptions)**
Sets the view descriptions representing the various editor associated with the editor support.

MultiViewDescription **getDefaultViewDescription()** Gets the view description representing the editor shown by default.

void setDefaultViewDescription(MultiViewDescription **defaultViewDescription)**
Sets the view description representing the editor shown by default.

4.3 org.mcphase.base.SourceEditor

org.mcphase.base.SourceEditor implements an editor providing access to a file using a plain text interface. Please see 'Rich Client Programming: Plugging into the NetBeans Platform' for a full description of the implementation.

4.4 org.mcphase.base.SourceView

org.mcphase.base.SourceView implements a view description for a plain text editor.

4.5 org.mcphase.base.ViewBase

org.mcphase.base.ViewBase implements a MultiViewDescription to streamline the development process.

It provides the following functionality

MultiViewElement **getViewElement()** Gets the element associated with the view (i.e. the editor).

void setViewElement(MultiViewElement **viewElement)** Sets the element associated with the view (i.e. the editor).

String **getDisplayName()** Gets the name displayed in the tab bar.

void setDisplayName(String **displayName)** Sets the name displayed in the tab bar.

Image getIcon() Gets the icon displayed next to the file name while the view is selected.

void setIcon(Image icon) Sets the icon displayed next to the file name while the view is selected.

4.6 org.mcphase.base.VisualEditorBase

org.mcphase.base.VisualEditorBase implements a basic visual editor, which can be embedded in a multiple-editor environment. Components may be registered with key-value type variables such that they are kept synchronised.

It provides the following functionality

void updateControls() Updates the content of all registered controls.

void registerComponent(String variable, Component component)
Registers the specified component with the specified variable such that they are kept synchronised. Supported components are JCheckBox, JComboBox, JTextComponent.

String getVariable(String name) Gets the value of the named variable or *null* if it cannot be found.

void setVariable(String variable, String value) Sets the value of the named variable or removes the variable if the value is *null*.